DIPTI DHAWADE

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LinkedIn GitHub Portfolio

EDUCATION

BTech - Computer Science and Artificial Intelligence, Plaksha University	8.58	*2027*
12th - Jawahar Navodaya Vidyalaya Ahmednagar	96.4%	*2021*
10th - Jawahar Navodaya Vidyalaya Ahmednagar	96.2%	*2019*

PROJECTS

NavBot | TypeScript, React, Node.js, Express, PostgreSQL, Pinecone, Docker *Jan 2026 - Present*

- Built a **full-stack SaaS** that adds a **site-grounded AI chatbot** to any website with one script tag, as a **TypeScript monorepo** with a **React** dashboard, **Node.js/Express REST APIs**, **PostgreSQL** & **OAuth**.
- Built a **Playwright crawler** with cron-based incremental re-indexing into **Pinecone** and an **agentic RAG pipeline** (**query planning, reranking, streaming**) scoring **4.8/5 groundedness** on **100 eval queries**.

Flagzen | React, Three.js, Konva, Node.js, Express, MongoDB *Nov 2025 - Oct 2026*

- Built a **3D merch customisation platform** where shoppers design on **3D product models** in the browser and order them.
- Engineered a seller **glTF (.glb) upload pipeline** with server-side validation and **click-to-place print areas** stored in model space; rendered the catalogue via a single shared **WebGL canvas** with scissored viewports.
- Developed a **Konva design studio** (layers, undo/redo, logo-driven layout suggestions, high-resolution print export) with a **live 3D view** that projects the design onto the model as a decal.

Interview.ai | React, Python, FastAPI, LlamaIndex, PostgreSQL, Docker *Sep 2025 - Oct 2026*

- Built a **full-stack interview-prep platform** that turns scattered **SDE prep into a self-paced 1-6 month plan**: a **265-topic roadmap** (DSA, System Design, OS, etc.), a **library of 138 course PDFs**, and an **AI tutor** that **answers from the student's uploaded notes** and **makes flashcards** and **quizzes** for spaced-repetition revision.
- Designed the **RAG pipeline** with **LlamaIndex on PostgreSQL + pgvector**, using **OCR for notes, hybrid vector + keyword search and cross-encoder re-ranking**. It runs on a **FastAPI backend in Docker**, with **GitHub Actions CI/CD** running **140 tests** on every change.

Mobile-Hi-SAM | Python, PyTorch *Aug 2025 - Dec 2025*

- Built an **edge-deployable text segmentation model** by replacing **Hi-SAM's 650M-parameter encoder** with **MobileSAM's TinyViT**, plus a **custom adapter** and **word/line/paragraph decoder**.
- **Retained ~62% of Hi-SAM's panoptic quality** with only **12.6M parameters**, a **~98% smaller model**.

EXPERIENCE

Indian School of Business, Hyderabad - Research Intern *Jun 2026 - Jul 2026*

- Developed a **multi-dimensional climatic vulnerability model** for rural India using **geospatial**, **climate**, and **government survey** datasets, covering ~7,000 blocks under Prof. Ashwini Chhatre.
- Python, Google Earth Engine (GEE), Data Analysis

Reimagining Higher Education Foundation, Plaksha University - Intern *Oct 2024 – May 2025*

- **Coordinated high-stakes events** like Founders' Day and Foundation Day at Plaksha University & conducted campus tours for founders, prospective students, and visitors. Researched and managed prospect donors list to support **fundraising**.

SKILLS

- **Programming Languages**: C, C++, Python (PyTorch), SQL, JavaScript, TypeScript, HTML/CSS
- **Web and Backend:** React.js, Node.js, Express.js, FastAPI, REST APIs, JWT/OAuth
- **Data, AI & Infrastructure:** PostgreSQL, MongoDB, Supabase, Pinecone, ChromaDB, RAG, Docker, GitHub Actions, Git
- **Core CS:** Data Structures & Algorithms, OOP, Operating Systems, DBMS, Computer Networks

LEADERSHIP & ACHIEVEMENTS

- **Ranked top 5** in the class in the courses **Machine Learning & Pattern Recognition** and **Design & Analysis of Algorithms (A grade)**; **A- grades** in courses **Foundations of Computer Systems, Theory of Computation, Computer Networks, and Deep Learning.**
- Secured **3rd place - SP Dutt Award for Innovation** by designing and building a soft robotic gripper for underwater manipulation.
- Member of the **DS Brar Centre for Girls and Women in STEM (GWiST),** Plaksha University: Promote gender equity in STEM.
- **Millennium Fellow 2024, MCN & United Nations:** Led a social impact project aligned with SDG 8: **Decent Work and Economic Growth** empowering rural and tribal communities toward sustainable livelihoods.
- **Lila Girl 2021:** Awarded by Lila Poonawalla Foundation for academic excellence and social impact.
- **Vice Head Girl at Jawahar Navodaya Vidyalaya:** Successfully managed the boarding program, fostered a strong school community of **500 students** and ensured student safety and well-being.
- Competed in **National Inter-School Sports & Games** against participants from **600+ schools** across India.